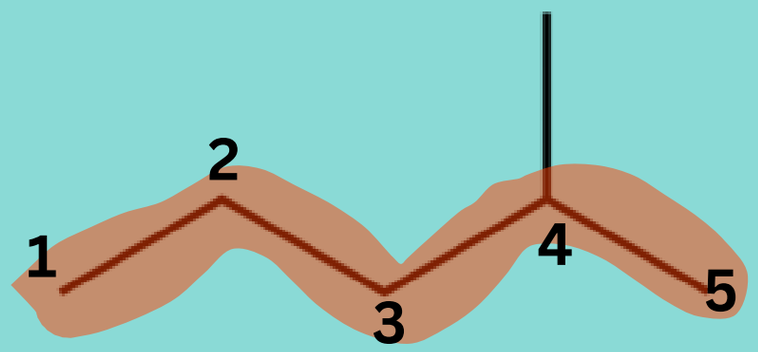
CARD A01 — STRUCTURE → NAME

Level: **FOUNDATION** Molecular Formula (counted): **C_6H_{14}**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain=5 C→ pentane. Two 5-C paths exist (through the right-hand CH_3 or through the upward branch), but each carries one methyl group, so they lead to the same name.

Numbering: left → right gives 4; right → left gives 2. First point of difference: 2 < 4, so number from the right.

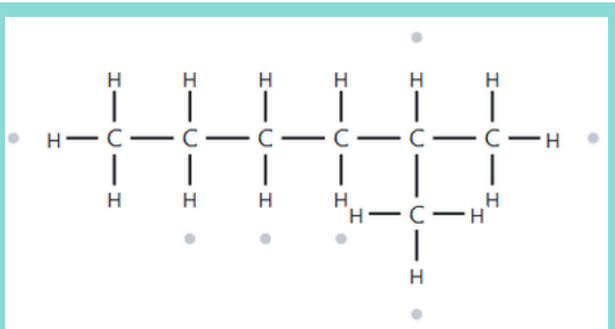
Substituents: methyl on C2 → 2-methyl

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK

✓ MATCH



2-methylhexane

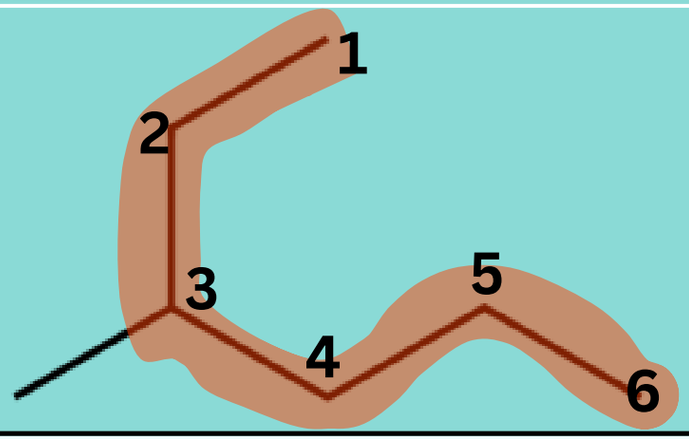
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A02 — STRUCTURE → NAME

Level: **FOUNDATION** Molecular Formula (counted): **C_7H_{16}**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 6 C → hexane (open chain). Only one 6-C path exists, running from the top line end down through the branched carbon and along to the bottom-right end. The path through the bottom-left line end is only 5 C, so that line is a branch.

Numbering: top end → bottom-right end gives 3; bottom-right end → top end gives 4. First point of difference: 3 < 4, so number from the top end.

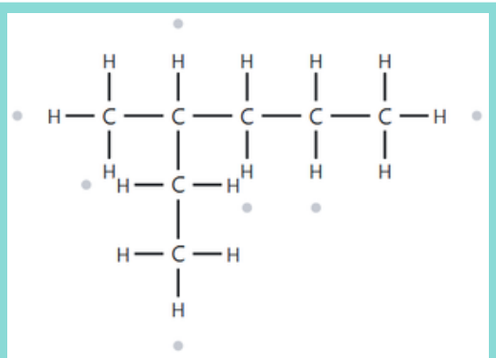
Substituents: methyl on C3 → 3-methyl

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK

✓ MATCH



3-methylhexane

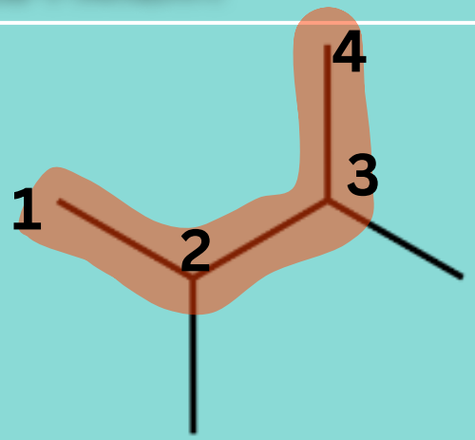
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A03 — STRUCTURE → NAME

Level: **FOUNDATION** Molecular Formula (counted): **C_6H_{14}**

1. NUMBERED PARENT



2. NAMING TRAIL

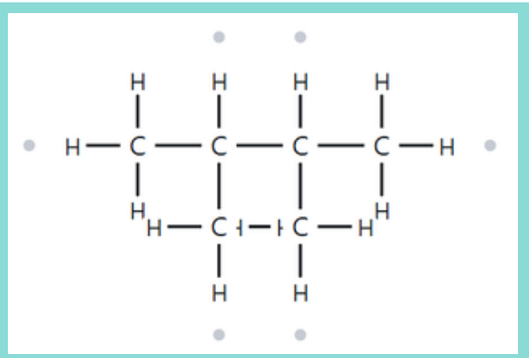
Parent: longest chain = 4 C → butane. Several 4-C paths exist, but each carries two methyl groups, so they lead to the same name.
Numbering: left → right gives 2,3; right → left gives 2,3. There is no point of difference, so the direction doesn't change the name and either one is fine.
Substituents: methyl on C2 and C3 → 2,3-dimethyl

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK

✓ MATCH



2,3-dimethylbutane

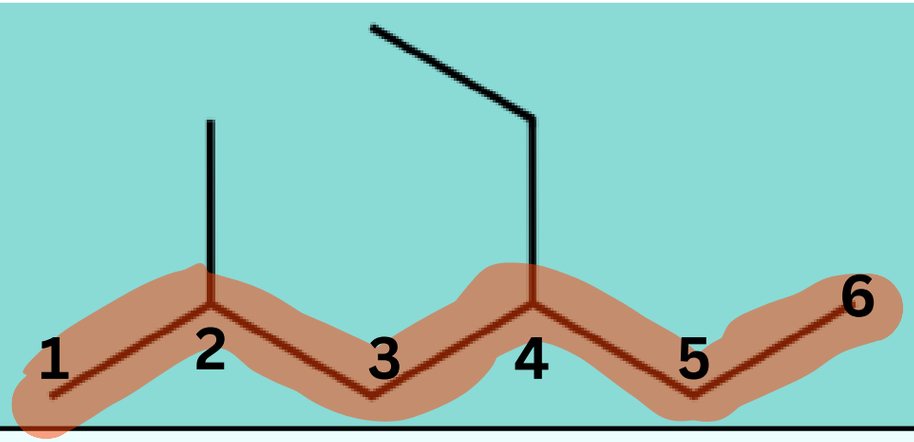
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A04 — STRUCTURE → NAME

Level: **DEVELOPING** Molecular Formula (counted): **C_9H_{20}**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 6 C → hexane. Several 6-C paths exist, but each carries one methyl and one ethyl group, so they lead to the same name.

Numbering: left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.

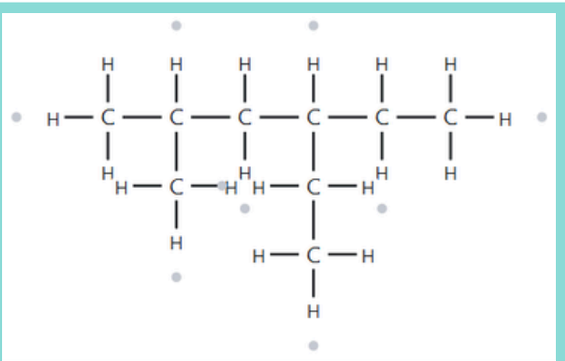
Substituents: ethyl on C4 and methyl on C2, listed alphabetically (e before m) → 4-ethyl-2-methyl.

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK

✓ MATCH



4-ethyl-2-methylhexane

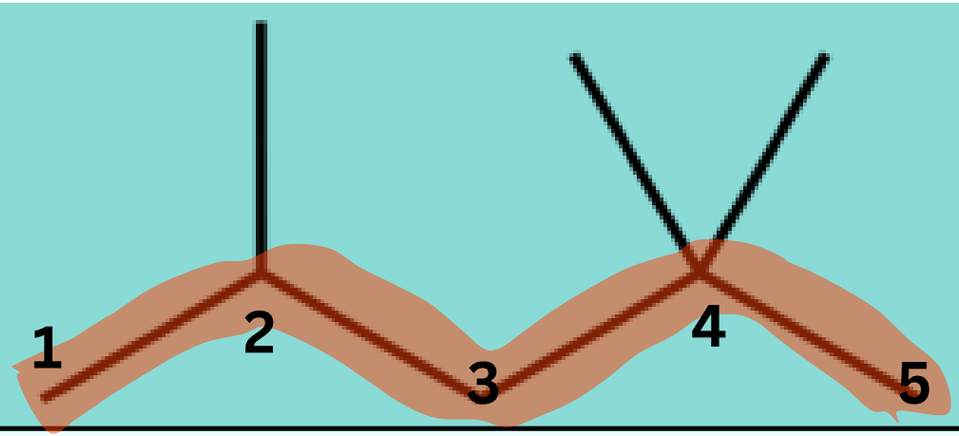
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A05 — STRUCTURE → NAME

Level: **DEVELOPING** Molecular Formula (counted): **C_8H_{18}**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 5 C → pentane. Several 5-C paths exist, but each carries three methyl groups, so they lead to the same name.

Numbering: left → right gives 2,4,4; right → left gives 2,2,4. First point of difference: 2 < 4, so number from the right.

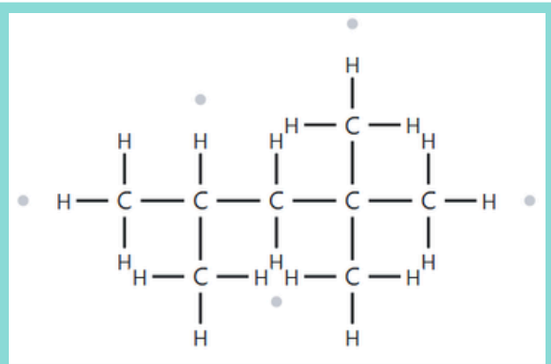
Substituents: methyl on C2, C2 and C4 → 2,2,4-trimethyl. The locant is repeated for each methyl on the same carbon.

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK

✓ MATCH



2,2,4-trimethylpentane

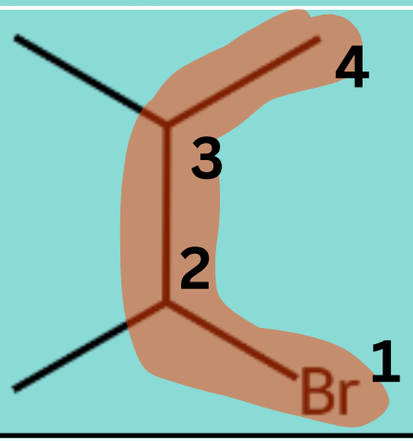
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A06 — STRUCTURE → NAME

Level: **DEVELOPING** Molecular Formula (counted): **$C_5H_{11}Br$**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 4 C → butane. Several 4-C paths exist, but each carries one bromo and one methyl group, so they lead to the same name.

Numbering: lower-right end → upper end gives 2,3; upper end → lower-right end gives 2,3. The locant sets are identical, so there is no first point of difference. The tie goes to alphabetical order: bromo comes before methyl, so the bromine gets the lower number, and I number from the lower-left end.

Substituents: bromo on C2 and methyl on C3, listed alphabetically (b before m) → 2-bromo-3-methyl.

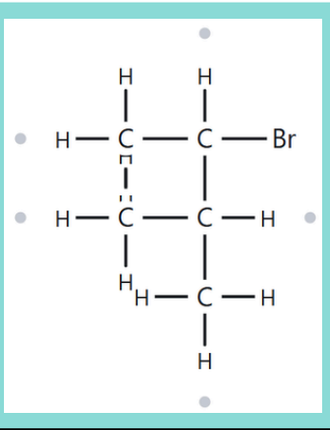
3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK

✓ MATCH

2-bromo-3-methylbutane



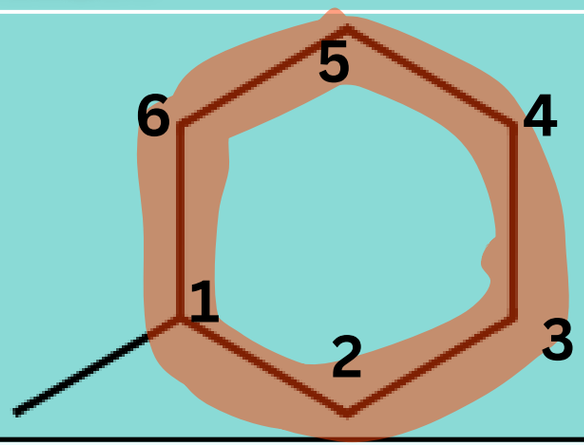
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A07 — STRUCTURE → NAME

Level: **DEVELOPING** Molecular Formula (counted): **C_7H_{14}**

1. NUMBERED PARENT



2. NAMING TRAIL

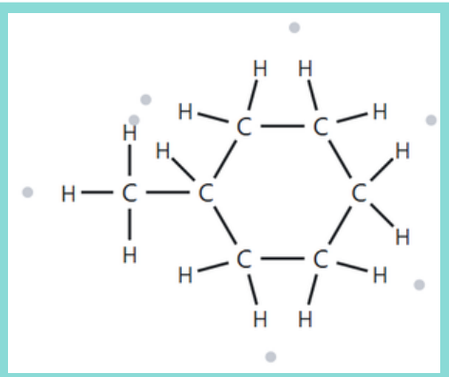
Parent: the ring has 6 C and the only chain attached has 1 C, so the ring is the parent → cyclohexane (ring, all single bonds).
Numbering: one substituent, so C1 is the carbon with the methyl. Both directions give {1}, so there's no point of difference.
Substituents: methyl on C1 → methyl

3. MY NAME

methylcyclohexane

4. THE ALKANE CHECK

✓ MATCH


methylcyclohexane

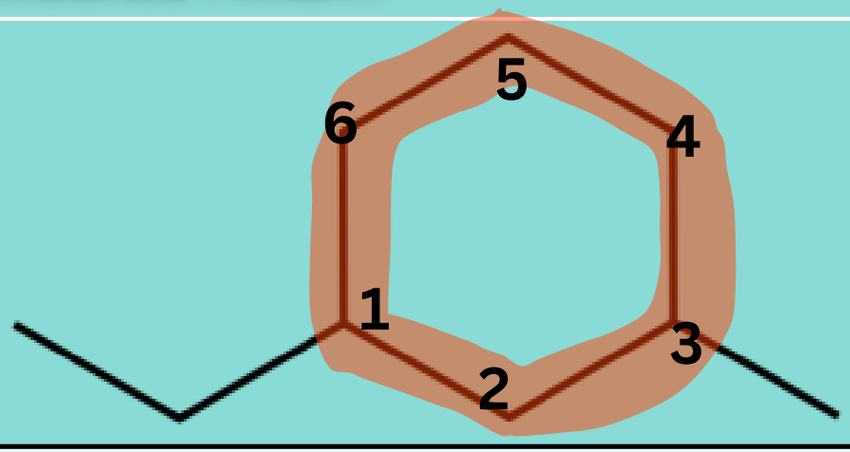
- VERIFICATION STRIP -

✓ Longest Chain Ring ✓ Lowest Locants ✓ Alphabetical Order ✓ Punctuation ✓ Formula Counted

CARD A08 — STRUCTURE → NAME

Level: **CHALLENGE** Molecular Formula (counted): **C_9H_{18}**

1. NUMBERED PARENT



2. NAMING TRAIL

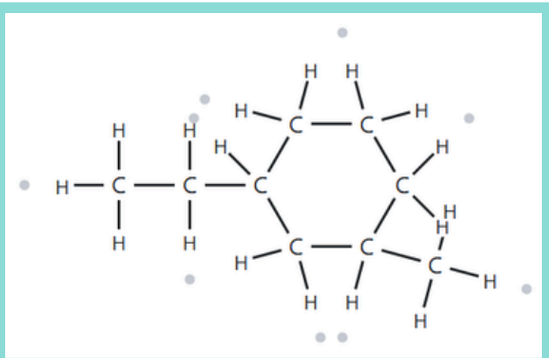
Parent: the ring has 6 C and the longest attached chain (the ethyl group) has 2 C, so the ring is the parent → cyclohexane (ring, all single bonds).
Numbering: both directions give 1,3. The tie goes to alphabetical order, so ethyl gets C1.
Substituents: ethyl on C1 and methyl on C3, listed alphabetically (e before m) → 1-ethyl-3-methyl

3. MY NAME

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK

✓ MATCH



1-ethyl-3-methylcyclohexane

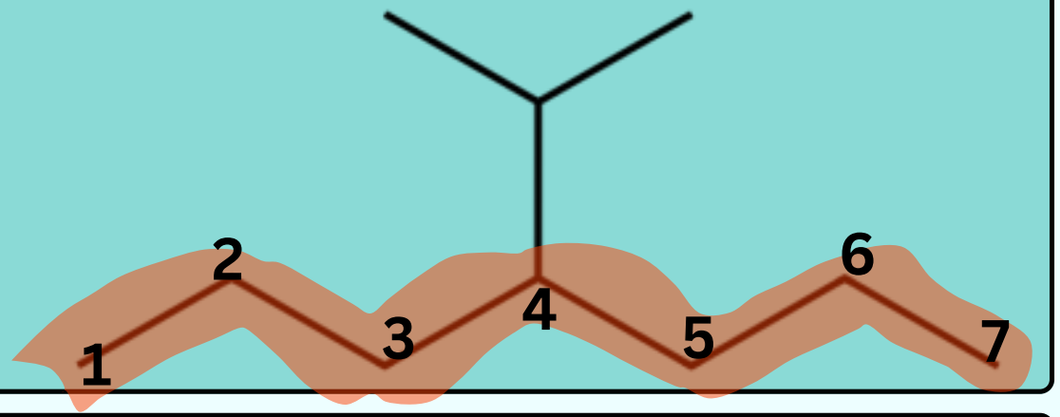
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A09 — STRUCTURE → NAME

Level: **CHALLENGE** Molecular Formula (counted): **$C_{10}H_{22}$**

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 7 C → heptane. The 7-C path runs along the whole zigzag from the left end to the right end. Any path through the upward branch gives only 6 C, so that branch is a substituent.

Numbering: left → right gives 4; right → left gives 4. The locant sets are identical, so there's no point of difference and either direction gives the same name.

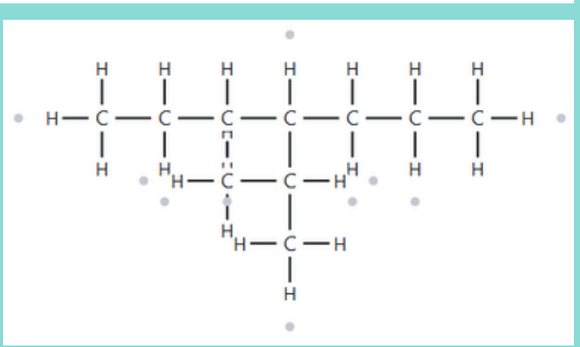
Substituents: isopropyl on C4 → 4-isopropyl

3. MY NAME

4-isopropylheptane

4. THE ALKANE CHECK

X NOT A MATCH



2-methyl-3-propylhexane

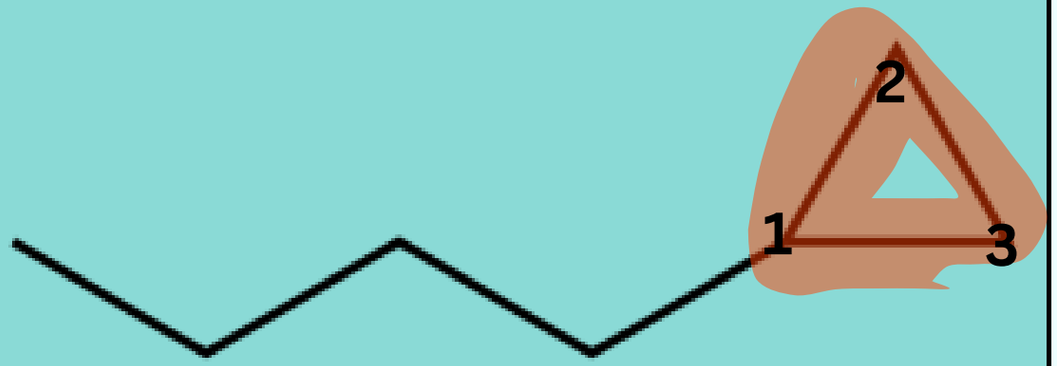
- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD A10 — STRUCTURE → NAME

Level: **CHALLENGE** Molecular Formula (counted): **C_7H_{14}**

1. NUMBERED PARENT



2. NAMING TRAIL

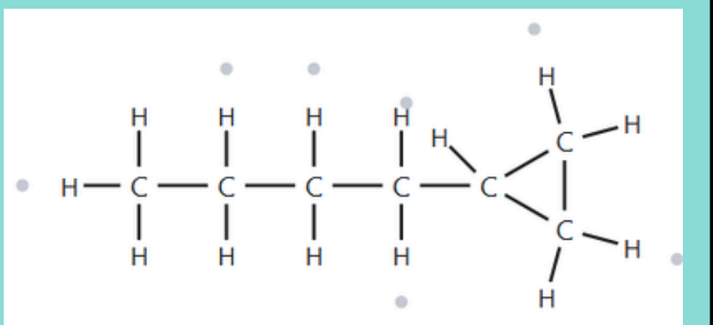
Parent: the ring has 3 C and the attached chain has 4 C. When the ring is the parent whenever a ring is present, so the parent is cyclopropane.
Numbering: one substituent, so C1 is the ring carbon that holds it. Both directions give {1}, so there's no point of difference and no locant is needed.
Substituents: butyl (4 C, unbranched) on C1 → butyl

3. MY NAME

butylcyclopropane

4. THE ALKANE CHECK

✓ MATCH



butylcyclopropane

- VERIFICATION STRIP -

- Longest Chain Ring
- Lowest Locants
- Alphabetical Order
- Punctuation
- Formula Counted

CARD B01 — NAME → STRUCTURE

Level: **FOUNDATION** Given Name: **2-methylpropane**

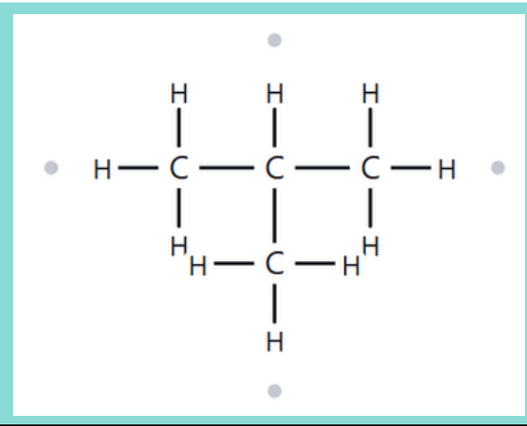
1. NAME DECODED

Parent: propane → a 3-carbon chain

Substituents: methyl on C2

Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_4H_{10}

Open Chain: C_nH_{2n+2} = C_4H_{10}. There are no halogens, so nothing is subtracted.

4. BACK-CHECK

Returned by the Alkane:

2-methylpropane

✓ Same as the given name

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded

CARD B02 — NAME → STRUCTURE

Level: **FOUNDATION** Given Name: **3-ethylpentane**

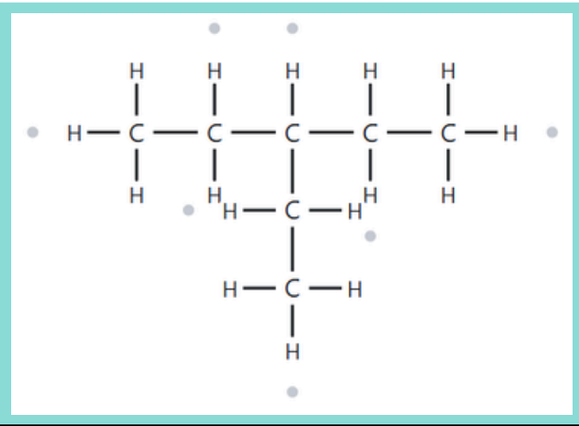
1. NAME DECODED

Parent: pentane → a 5-carbon chain

Substituents: ethyl on C3

Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_7H_{16}

Open Chain: C_nH_{2n+2} = C_7H_{16}. There are no halogens, so nothing is subtracted.

4. BACK-CHECK

Returned by the Alkane:

3-ethylpentane

✓ Same as the given name

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded

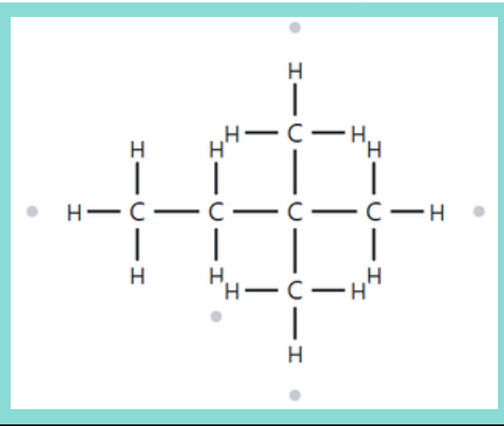
CARD B03 — NAME → STRUCTURE

Level: **FOUNDATION** Given Name: **2,2-dimethylbutane**

1. NAME DECODED

Parent: butane → a 4-carbon chain.
Substituents: methyl on C2; methyl on C2
Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_6H_{14}
Open Chain: C_nH_{2n+2} = C_6H_{14}. There are no halogens, so nothing is subtracted.

4. BACK-CHECK

Returned by the Alkane:

2,2-dimethylbutane

✓ Same as the given name

- VERIFICATION STRIP -

- Parent Length Matches
- Substituent Positions Match
- Formula Agrees
- Back-Check Recorded

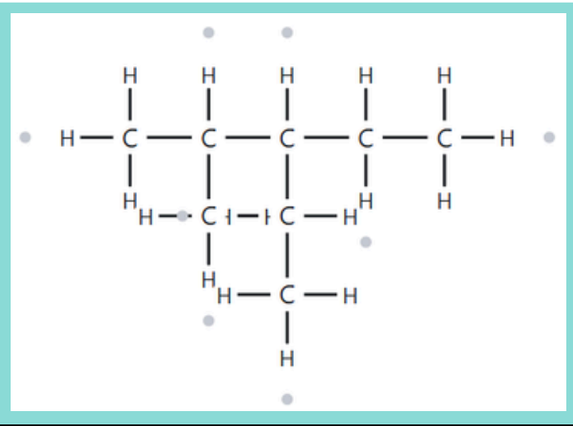
CARD B04 — NAME → STRUCTURE

Level: **DEVELOPING** Given Name: **3-ethyl-2-methylpentane**

1. NAME DECODED

Parent: pentane → a 5-carbon chain
Substituents: methyl on C2; ethyl on C3
Expected formula: 5 + 1 + 2 = 8 C; open chain → C_8H_{18}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_8H_{18}
Open Chain: C_nH_{2n+2} = C_8H_{18}. There are no halogens, so nothing is subtracted.

4. BACK-CHECK

Returned by the Alkane:
3-ethyl-2-methylpentane

✓ **Same as the given name**

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded

CARD B05 — NAME → STRUCTURE

Level: **DEVELOPING** Given Name: **1-chloro-2-methylpropane**

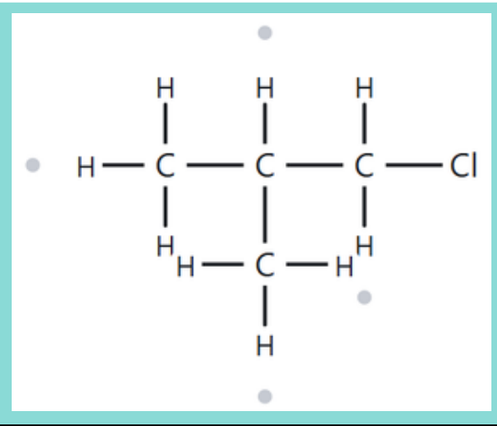
1. NAME DECODED

Parent: propane → a 3-carbon open chain

Substituents: chloro on C1; methyl on C2

Expected formula: 3 + 1 = 4 C; open chain (alkane) → C_4H_{10}; Cl replaces one H → C_4H_9Cl

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing:

C_4H_9Cl

Open Chain: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl

4. BACK-CHECK

Returned by the Alkane:

1-chloro-2-methylpropane

✓ **Same as the given name**

- VERIFICATION STRIP -

- Parent Length Matches
- Substituent Positions Match
- Formula Agrees
- Back-Check Recorded

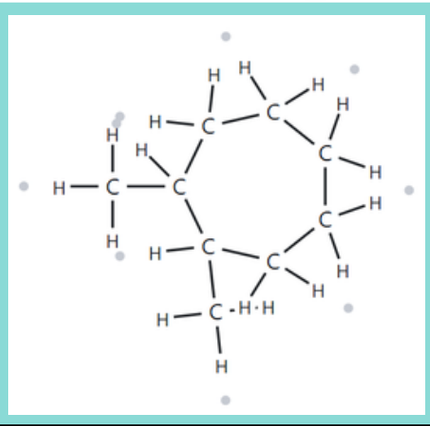
CARD B06 — NAME → STRUCTURE

Level: **DEVELOPING** Given Name: **1,2-dimethylcyclopentane**

1. NAME DECODED

Parent: cyclopentane → a 5-carbon ring
Substituents: methyl on C1; methyl on C2
Expected formula: 5 + 1 + 1 = 7 C; one ring → C_nH_{2n} = C_7H_{14}. There are no halogens, so nothing is subtracted. The expected formula is C_7H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_7H_{14}
Open Chain: One ring: C_nH_{2n} = C_7H_{14}. No halogens, so no H is replaced. The counted formula matches the expected formula.

4. BACK-CHECK

Returned by the Alkane:
1,2-dimethylcyclopentane

✓ **Same as the given name**

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded

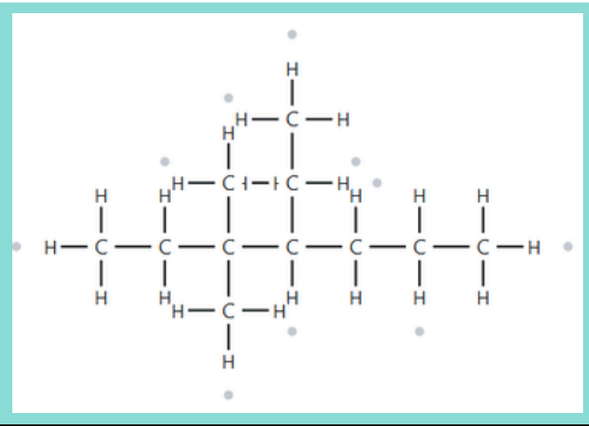
CARD B07 — NAME → STRUCTURE

Level: **DEVELOPING** Given Name: **4-ethyl-3,3-dimethylheptane**

1. NAME DECODED

Parent: heptane, a 7-carbon open chain
Substituents: methyl on C1; methyl on C2
Expected formula: 5 ring carbons + 1 + 1 = 7 C. One ring means C_nH_{2n}, so C_7H_{14}. There are no halogens, so no H is replaced

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$
Open Chain: C_nH_{2n+2} = $C_{11}H_{(2\times11+2)}$ = $C_{11}H_{24}$. The counted formula matches the general formula, so the drawing is consistent with the name.

4. BACK-CHECK

Returned by the Alkane:

4-ethyl-3,3-dimethylheptane

✓ **Same as the given name**

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded

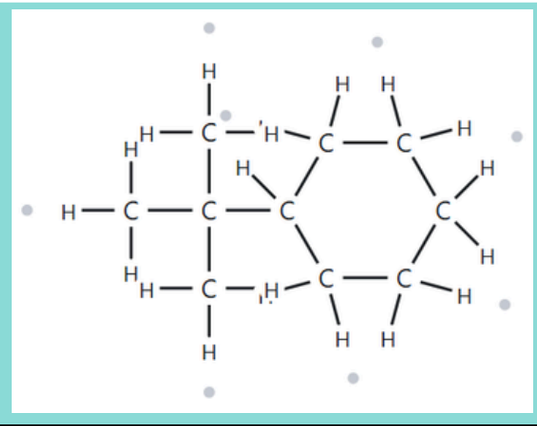
CARD B08 — NAME → STRUCTURE

Level: **CHALLENGE** Given Name: **tert-butylcyclohexane**

1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring
Substituents: tert-butyl on C1. A single substituent on a ring, so the locant "1" is not written in the name. The tert-butyl group is $-C(CH_3)_3$, which has 4 carbons.
Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$; the substituent replaces one H on the ring, and the substituent is itself an alkyl group.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
Open Chain: C_nH_{2n+2} = $C_{10}H_{20}$. No halogens, so nothing is subtracted. The drawing matches the formula.

4. BACK-CHECK

Returned by the Alkane:
(1,1-dimethylethyl)cyclohexane

X Not the same as the given name

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded

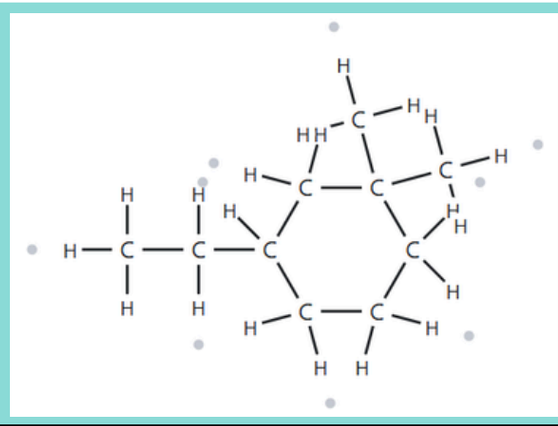
CARD B09 — NAME → STRUCTURE

Level: **CHALLENGE** Given Name: **3-ethyl-1,1-dimethylcyclohexane**

1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.
Substituents: two methyl groups on C1 (both on the same carbon); ethyl on C3
Expected formula: 6 (ring) + 2 (two methyls) + 2 (ethyl) = 10 C; one ring → C_nH_{2n} = $C_{10}H_{20}$. There are no halogens, so no H is removed.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$
One ring: C_nH_{2n} = $C_{10}H_{20}$ ✓ (the count matches the general formula, so the drawing is consistent with the name).

4. BACK-CHECK

Returned by the Alkane:

3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded

CARD B10 — NAME → STRUCTURE

Level: **CHALLENGE** Given Name: **2-ethylbutane**

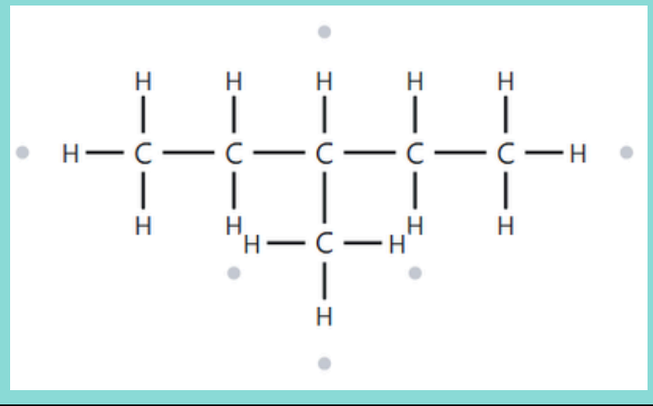
1. NAME DECODED

Parent: butane → a 4-carbon open chain

Substituents: ethyl on C2

Expected formula: 4 + 2 = 6 C; open chain → $C_nH_{2n} + 2 = C_6H_{14}$.

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_6H_{14}

One ring: $C_nH_{2n} + 2 = C_6H_{14}$. There are no halogens, so no H is replaced.

4. BACK-CHECK

Returned by the Alkane:

3-methylpentane

X Not the same as the given name

- VERIFICATION STRIP -

- ✓ Parent Length Matches
- ✓ Substituent Positions Match
- ✓ Formula Agrees
- ✓ Back-Check Recorded